

**NASA MAST Archive
(TESS FITS files)**

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graph TD; A[NASA MAST Archive (TESS FITS files)] --> B[STEP 1-2: Data Collection  
Download + Preprocess  
1146 stars -> cleaned .npy]; B --> C[STEP 3: BLS Detection  
Period search 0.3-27 days  
217 candidates found]; C --> D[STEP 4: Phase Folding  
Stack transits -> 200-pt curve]; D --> E[STEP 5: ML Classification  
Random Forest, 34 features  
89% accuracy]; E --> F[STEP 6: Parameter Estimation  
Period, Depth, Duration, SNR]; F --> G[OUTPUT  
3 Planets | 185 EBs | 18 Blends  
TIC 25858367 — New Finding];
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